

NCT07563153: Impact of Optimized Pacing Strategies on Clinical and Hemodynamic Outcomes in Heart Failure Patients With Pacemaker

Gender: ALL

Age: 19 Years - MISSING

Criteria:

Inclusion Criteria:

* Patients with symptomatic bradycardia who meet the indication for permanent pacemaker implantation and fulfill one of the following conditions:

1. Sick sinus syndrome with or without impaired atrioventricular conduction
2. Persistent or permanent atrial fibrillation with slow ventricular response
3. Chronotropic incompetence

* Patients diagnosed with heart failure with left ventricular ejection fraction $\geq 50\%$ on transthoracic echocardiography with at least one of the following:

* H2FPEF score ≥ 6 or HFA-PEFF score ≥ 5

* N-terminal pro-B-type natriuretic peptide ≥ 300 pg/mL (sinus rhythm) or ≥ 600 pg/mL (atrial fibrillation)

* Prior hospitalization for heart failure or documented use of loop diuretics for heart failure symptoms

Exclusion Criteria:

* Patients expected to have a ventricular pacing burden $\geq 20\%$ without sufficient capture of cardiac physiologic pacing, which includes biventricular pacing, His bundle pacing, and left bundle branch area pacing.

(Sufficient cardiac physiologic pacing is defined as a paced QRS duration ≤ 140 ms.)

* Patients not expected to achieve sufficient pacing dependency, defined as:

1. In sinus rhythm: baseline atrial rate ≥ 60 bpm on Holter monitoring or inpatient ECG monitoring
2. In atrial fibrillation/flutter: baseline ventricular rate ≥ 60 bpm on Holter monitoring or inpatient ECG monitoring

* Patients with contraindications to permanent pacemaker implantation

* Patients with moderate or greater valvular stenosis or regurgitation.

* Patients with dyspnea not attributable to heart failure, due to uncontrolled comorbid conditions

* Pregnant or breastfeeding women.

* Patients who have refused active treatment.